Vittoria Ceccatelli

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EDUCATION

Sep 2025 – Mar 2026	Master Thesis at Harvard Medical School: "A Standardized Pipeline for Automated Carotid Artery Analysis" *Presented at the Harvard Surgery Research Day*
2023 – 2026	*MSc ETH Zurich in Biomedical Engineering: 5.43* *Relevant Coursework: Image Analysis and Computer Vision; Machine Learning; Mobile Health and Activity Monitoring; Product Design in Medical Engineering; Foundations of Materials Science I & II; Sports Biomechanics*
2023	Exchange semester at *National University of Singapore (NUS)* *Relevant Coursework: Robotics in Rehabilitation; Genomic Data Analysis*
2022 – 2026	Member of the *Swiss Study Foundation* selected based on high academic performance (Switzerland)
2020 – 2023	*BSc ETH Zurich in Health Sciences and Technology*: 5.32 (top 15%)
2020: *Dual Diploma*	German Abitur (High School Diploma): 1.4, International Baccalaureate (IB): 38 points

TECHNICAL SKILLS

Programming	Python, MATLAB, HTML, CSS, JavaScript, Git, Linux, R, SQL, Bash
Machine/Deep Learning	PyTorch, PyTorch Forecasting, TensorFlow, Supervised Learning, XGBoost, Validation and Evaluation
Data and Signals	Feature Engineering, Class Imbalance Strategies, Data Cleaning, VTK, Slicer
Other	CAD, CFD, Medical Image Segmentation, scikit-learn, Geomagic Wrap and DesignX, PTC Creo, ANSYS

EXPERIENCE AND RESEARCH

Sep 2025 – Mar 2026 — Applied ML/Data Engineering Pipeline Project in the Nezami Lab (Harvard Medical School, BWH, MGB)

- *Built a fully automated end-to-end ML pipeline using 111 raw CTA scans (DICOM) as inputs and delivering standardized multi-modal feature datasets with a 65.8% end-to-end success rate*
- *Engineered a modular feature extraction framework generating 354 features per patient spanning morphological, calcification, and hemodynamic domains, structured for use in downstream predictive modeling and large-cohort studies*
- *Integrated deep learning segmentation with automated geometry processing, 3D CFD simulation across 2 configurable solvers and 3 outlet boundary conditions, enabling deployment across varied computational environments*

Feb 2025 – Jun 2025 — Machine Learning Project on Wearable Data and Glucose Prediction (ETH Zurich, Switzerland)

- *Implemented end-to-end ML pipelines for three prediction tasks (classification, regression, and time-series forecasting) on multi-modal wearable data*
- *Performed feature engineering and selection across 69 derived features, implementing strategies to handle missing data, class imbalance, and inter-patient variability*
- *Compared baseline models and deep learning architectures, evaluating trade-offs between performance, robustness, and interpretability to inform practical modeling choices*

Feb 2024 – Mar 2024 — Data Analysis project in the Laboratory of Movement Biomechanics on the subject of *Mitigating fall risk in older adults using auditory noise stimulation* (ETH Zurich, Switzerland)

- *Advanced signal processing and statistical analysis on high-frequency EMG and force-plate (CoP) time-series data from controlled biomechanical experiments*
- *Developed MATLAB pipelines for EMG–EMG and EMG–CoP coherence analysis, statistical testing, and data visualization*

Oct 2023 – Jan 2024 — Intern in the R&D department of *Adler Ortho* designing acetabular cups (Milan, Italy)

- *Designed over 50 patient-specific hip implants using CT-based CAD tools (Geomagic Wrap and DesignX), supporting surgical planning and improving anatomical fit for orthopaedic procedures*
- *Worked cross-functionally with surgeons and QA teams to refine standard designs under regulatory constraints (PTC Creo)*
- *Created technical drawings and 3D visualizations for surgical planning and regulatory certification of orthopaedic devices*

Sep 2022 – Jun 2025 — Teaching Assistant in Programming and Data Analysis Courses (ETH Zurich, Switzerland)

- *Led one-on-one technical reviews of Python and MATLAB projects, focusing on code quality, efficiency, and structure*
- *Translated complex technical concepts for non-technical and interdisciplinary people*
- *Reviewed and evaluated over 40 projects per semester, providing structured and actionable feedback*

FURTHER COURSES

2023	*JavaScript Basics* Course on Coursera (UCDavis)
2022	*HTML, CSS, and Javascript for Web Developers* on Coursera (Johns Hopkins)
2022	Didactic Training *Coaching Students* for Teaching Assistants at ETH Zurich
2022	Innosuisse STARTUP CAMPUS Business Concept course (Zurich, Switzerland)

LANGUAGES

Italian (native), German (native), English (native), French (Level B2), Spanish (Level B1)

EXTRACURRICULAR INTERESTS

- Motivated long-distance runner, currently training for my third half marathon
- Pianist (ABRSM Grade 7 qualification)
- Keen volleyball player, participated in international youth tournaments and came second in the ISSA Volleyball Tournament 2017
- Certified ski instructor since November 2020, teaching children and adults (St. Moritz, Switzerland)